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INTRO TO LOG ANALYSIS WITH TRY HACK ME

1.1 Introduction

Log analysis is a critical component of cybersecurity and system monitoring. At a high level, it involves examining and interpreting event data generated by various sources—such as devices, applications, and systems—to monitor operational metrics and identify potential security incidents. The process includes collecting, parsing, and processing log files to transform raw data into actionable insights. An effective log analysis strategy enables security teams to respond more accurately to incidents and proactively detect potential threats.

2.1 Log Analysis Basics

Logs are recorded events or transactions within a system, device, or application.

Types of Logs

These log types include, but are not limited to:

- Application Logs: Messages from specific applications, providing insights into their status, errors, warnings, and other operational details.
- Audit Logs: Events, actions, and changes occurring within a system or application, providing a history of user activities and system behavior.
- Security Logs: Security-related events like logins, permission alterations, firewall activities, and other actions impacting system security.
- Server Logs: System logs, event logs, error logs, and access logs, each offering distinct information about server operations.
- System Logs: Kernel activities, system errors, boot sequences, and hardware status, aiding in diagnosing system issues.
- Network Logs: Communication and activity within a network, capturing information about events, connections, and data transfers.
- Database Logs: Activities within a database system, such as queries performed, actions, and updates.
- Web Server Logs: Requests processed by web servers, including URLs, source IP addresses, request types, response codes, and more.

I understand the basics of logs and I'm ready to proceed!

No answer needed

✓ Correct Answer

3.1 Investigation Theory

What's the term for a consolidated chronological view of logged events from diverse sources, often used in log analysis and digital forensics?

Answer: **Super Timeline**

Which threat intelligence indicator would `5b31f93c09ad1d065c0491b764d04933` and `763f8bdbc98d105a8e82f36157e98bbe` be classified as?

Answer: **File Hashes**

Answer the questions below

What's the term for a consolidated chronological view of logged events from diverse sources, often used in log analysis and digital forensics?

Super Timeline

✓ Correct Answer

Which threat intelligence indicator would `5b31f93c09ad1d065c0491b764d04933` and `763f8bdbc98d105a8e82f36157e98bbe` be classified as?

File Hashes

✓ Correct Answer

Task 4 ○ Detection Engineering

Common Log File Locations

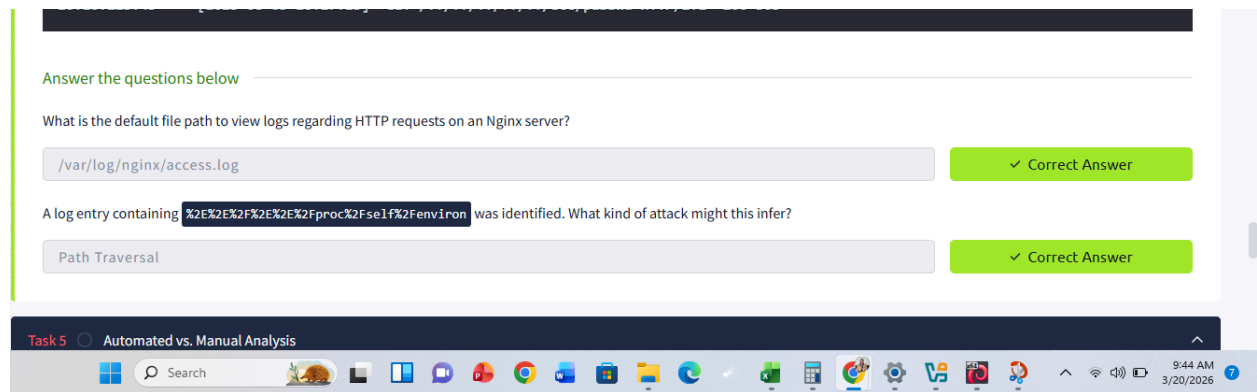
4.1 Detection Engineering

What is the default file path to view logs regarding HTTP requests on an Nginx server?

Answer: **/var/log/nginx/access.log**

A log entry containing `%2E%2E%2F%2E%2E%2Fproc%2Fself%2Fenviron` was identified. What kind of attack might this infer?

Answer: **Path Traversal**



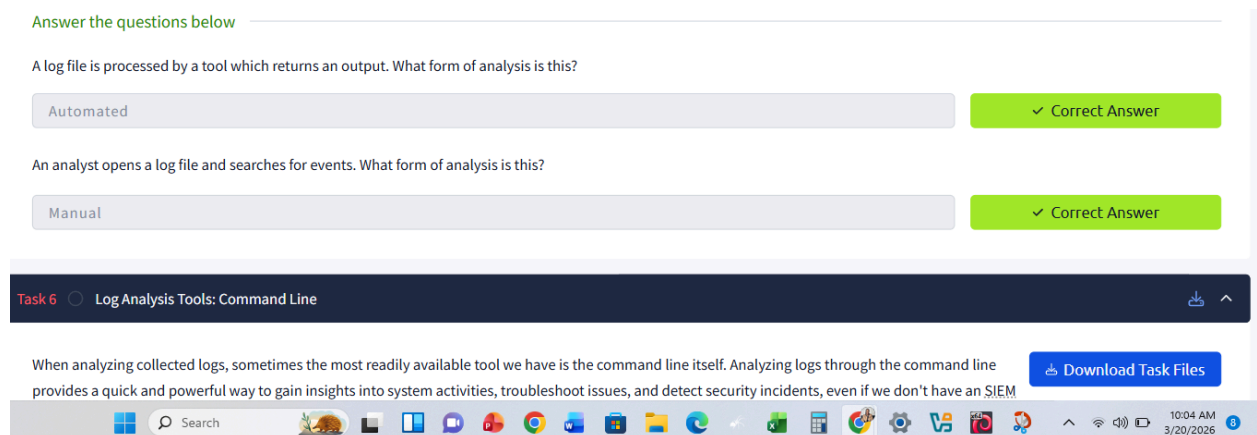
5.1 Automatic versus Manual Analysis

A log file is processed by a tool which returns an output. What form of analysis is this?

Answer: **Automated**

An analyst opens a log file and searches for events. What form of analysis is this?

Answer: **Manual**

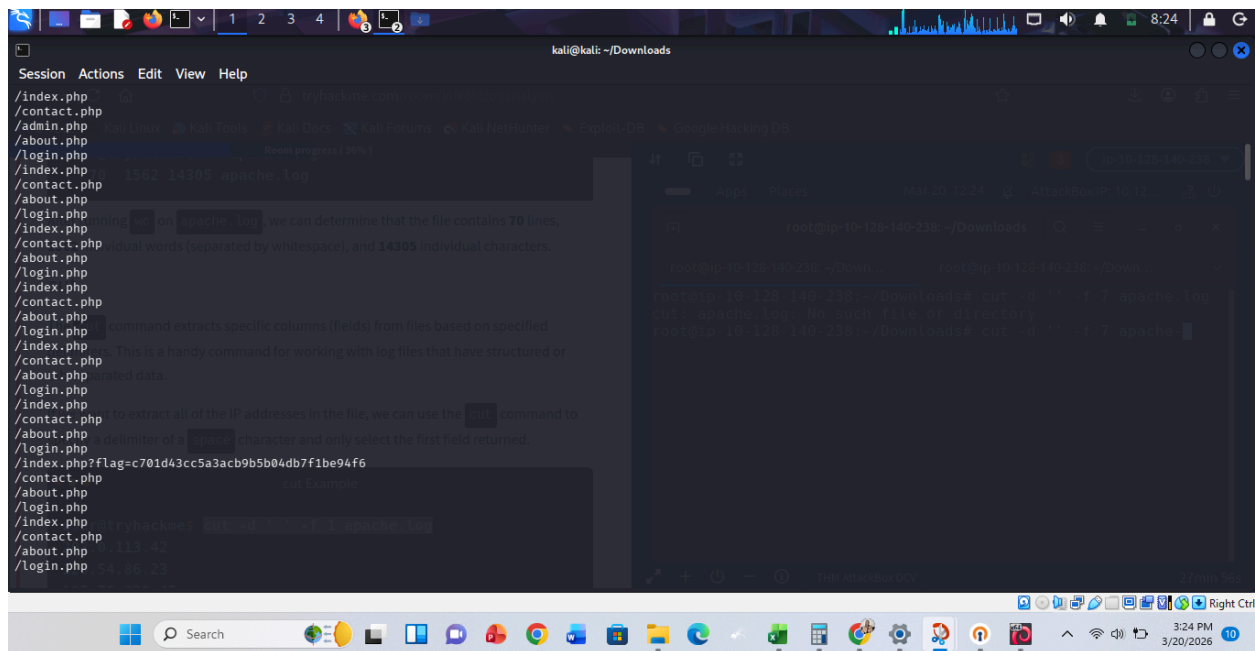


6.1 Log analysis tools:command line

a) Use `cut` on the `apache.log` file to return only the URLs. What is the flag that is returned in one of the unique entries?

Answer: **c701d43cc5a3acb9b5b04db7f1be94f6**

After downloading the task file called `apache.log`, I used `cut` to get the flag as shown in the screenshot below



Answer:52

```
File Machine view input Devices Help
kali@kali: ~/Downloads
Session Actions Edit View Help
203.64.78.90 - - [31/Jul/2023:12:34:32 +0000] "GET /login.php HTTP/1.1" 404 5678 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.330 Safari/537.36"
176.145.201.99 - - [31/Jul/2023:12:34:31 +0000] "GET /index.php?flag=c701d43cc5a3acb9b5b04db7f1be94f6 HTTP/1.1" 200 1234 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.270 Safari/537.36"
104.76.29.88 - - [31/Jul/2023:12:34:30 +0000] "GET /contact.php HTTP/1.1" 200 5678 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.350 Safari/537.36"
128.45.76.66 - - [31/Jul/2023:12:34:29 +0000] "GET /about.php HTTP/1.1" 404 4321 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.210 Safari/537.36"
76.89.54.221 - - [31/Jul/2023:12:34:28 +0000] "GET /login.php HTTP/1.1" 200 7890 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.330 Safari/537.36"
145.76.33.201 - - [31/Jul/2023:12:34:27 +0000] "GET /index.php HTTP/1.1" 200 4321 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.330 Safari/537.36"
110.122.65.76 - - [31/Jul/2023:12:34:26 +0000] "GET /contact.php HTTP/1.1" 200 7890 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.350 Safari/537.36"
203.64.78.90 - - [31/Jul/2023:12:34:25 +0000] "GET /about.php HTTP/1.1" 404 4321 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.81 Safari/537.36"
176.145.201.99 - - [31/Jul/2023:12:34:24 +0000] "GET /login.php HTTP/1.1" 200 1234 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.90 Safari/537.36"
104.76.29.88 - - [31/Jul/2023:12:34:23 +0000] "GET /index.php HTTP/1.1" 200 5678 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.81 Safari/537.36"
128.45.76.66 - - [31/Jul/2023:12:34:22 +0000] "GET /contact.php HTTP/1.1" 404 5678 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.54 Safari/537.36"
76.89.54.221 - - [31/Jul/2023:12:34:21 +0000] "GET /about.php HTTP/1.1" 200 1234 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.85 Safari/537.36"
145.76.33.201 - - [31/Jul/2023:12:34:20 +0000] "GET /login.php HTTP/1.1" 200 5678 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.90 Safari/537.36"

(kali@kali) ~/Downloads
$ awk '$9 == "200"' apache.log | wc -l
52

(kali@kali) ~/Downloads
$
```

c) In the apache.log file, which IP address generated the most traffic?

Answer: **145.76.33.201**

```
File Machine view input Devices Help
kali@kali: ~/Downloads
Session Actions Edit View Help
(kali@kali) ~/Downloads
$ cut -d ' ' -f1 apache.log | sort | uniq -c | sort -nr | head -n 1
8 145.76.33.201

(kali@kali) ~/Downloads
$
```

Sorry. We're having trouble getting your pages back.

We are having trouble restoring your last browsing session. Select Restore Session to try again.

Still not able to restore your session? Sometimes a tab is causing the issue. View previous tabs, remove the checkmarks from the tabs you don't need to recover, and then restore.

View Previous Tabs

Start New Session Restore Session

d) What is the complete timestamp of the entry where 110.122.65.76 accessed /login.php?

Answer:31/Jul/2023:12:34:40 +0000

The screenshot shows a Kali Linux terminal window with the following commands and output:

```
(kali@kali)-[~/Downloads]
$ cut -d ' ' -f1 apache.log | sort | uniq -c | sort -nr | head -n 1
      8 145.76.33.201

(kali@kali)-[~/Downloads]
$ grep "110.122.65.76" apache.log | grep "/login.php"
110.122.65.76 - - [31/Jul/2023:12:34:40 +0000] "GET /login.php HTTP/1.1" 200 9876 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/95.0.4638.110 Safari/537.36"
```

Below the terminal, a web browser window displays an error message: "Sorry. We're having trouble getting your pages back." It also includes a "Start New Session" button and a "Restore Session" button.

Answer the questions below

Use `cut` on the `apache.log` file to return only the URLs. What is the flag that is returned in one of the unique entries?

c701d43cc5a3acb9b5b04db7f1be94f6

✓ Correct Answer

In the `apache.log` file, how many total HTTP 200 responses were logged?

52

✓ Correct Answer

In the `apache.log` file, which IP address generated the most traffic?

145.76.33.201

✓ Correct Answer

What is the complete timestamp of the entry where `110.122.65.76` accessed `/login.php`?

31/Jul/2023:12:34:40 +0000

✓ Correct Answer

7.1 Log analysis tools:Regular expressions

a)How would you modify the original `grep` pattern above to match blog posts with an ID between 20-29?

Answer:post=2[0-9]

```
kali@kali: ~/Downloads/regex
Session Actions Edit View Help
grep: apache-ex2.log: No such file or directory

(kali@kali)~$ cd regex
cd: no such file or directory: regex

(kali@kali)~$ cd Downloads
cd: no such file or directory: Downloads

(kali@kali)~$ cd Downloads
(kali@kali)~/Downloads$ cd regex
(kali@kali)~/Downloads/regex$ grep -E 'post=2[0-9]' apache-ex2.log
66.78.92.112 - - [03/Aug/2023:05:21:55 +0000] "GET /blog.php?post=26 HTTP/1.1" 200 - "Mozilla/5.0"
187.22.99.57 - - [04/Aug/2023:23:40:10 +0000] "GET /blog.php?post=29 HTTP/1.1" 200 - "Mozilla/5.0"
202.67.54.90 - - [06/Aug/2023:23:38:25 +0000] "GET /blog.php?post=28 HTTP/1.1" 200 - "Mozilla/5.0"
75.102.56.44 - - [08/Aug/2023:15:23:59 +0000] "GET /blog.php?post=22 HTTP/1.1" 200 - "Mozilla/5.0"
109.54.87.56 - - [08/Aug/2023:21:49:39 +0000] "GET /blog.php?post=24 HTTP/1.1" 200 - "Mozilla/5.0"
45.98.99.54 - - [09/Aug/2023:15:23:04 +0000] "GET /blog.php?post=23 HTTP/1.1" 200 - "Mozilla/5.0"
102.210.56.99 - - [12/Aug/2023:05:18:46 +0000] "GET /blog.php?post=27 HTTP/1.1" 200 - "Mozilla/5.0"
210.56.102.76 - - [13/Aug/2023:05:18:18 +0000] "GET /blog.php?post=20 HTTP/1.1" 200 - "Mozilla/5.0"
76.44.78.210 - - [15/Aug/2023:15:20:16 +0000] "GET /blog.php?post=28 HTTP/1.1" 200 - "Mozilla/5.0"
76.56.102.210 - - [16/Aug/2023:05:16:18 +0000] "GET /blog.php?post=25 HTTP/1.1" 200 - "Mozilla/5.0"
76.210.56.102 - - [17/Aug/2023:05:15:43 +0000] "GET /blog.php?post=23 HTTP/1.1" 200 - "Mozilla/5.0"

(kali@kali)~/Downloads/regex$
```

b)What is the name of the filter plugin used in Logstash to parse unstructured log data?

Answer:Grok

Answer the questions below

How would you modify the original `grep` pattern above to match blog posts with an ID between 20-29?

✓ Correct Answer

What is the name of the filter plugin used in Logstash to parse unstructured log data?

✓ Correct Answer

Task 8 Log Analysis Tools: CyberChef

CyberChef is a powerful tool in an analyst's toolkit. Created by GCHQ, CyberChef has long been regarded as the "Cyber Swiss Army Knife." The application

[Download Task Files](#)

8.1 Log analysis tools:CyberChef

a)Locate the "loganalysis.zip" file under `/root/Rooms/introloganalysis/task8` and extract the contents.

Answer:No need to answer

b) Upload the log file named "access.log" to CyberChef. Use regex to list all of the IP addresses. What is the full IP address beginning in 212?

Answer:212.14.17.145

Explanation: i used this syntax `grep -oE "212\.[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}" access.log | sort -u`

[illegible]

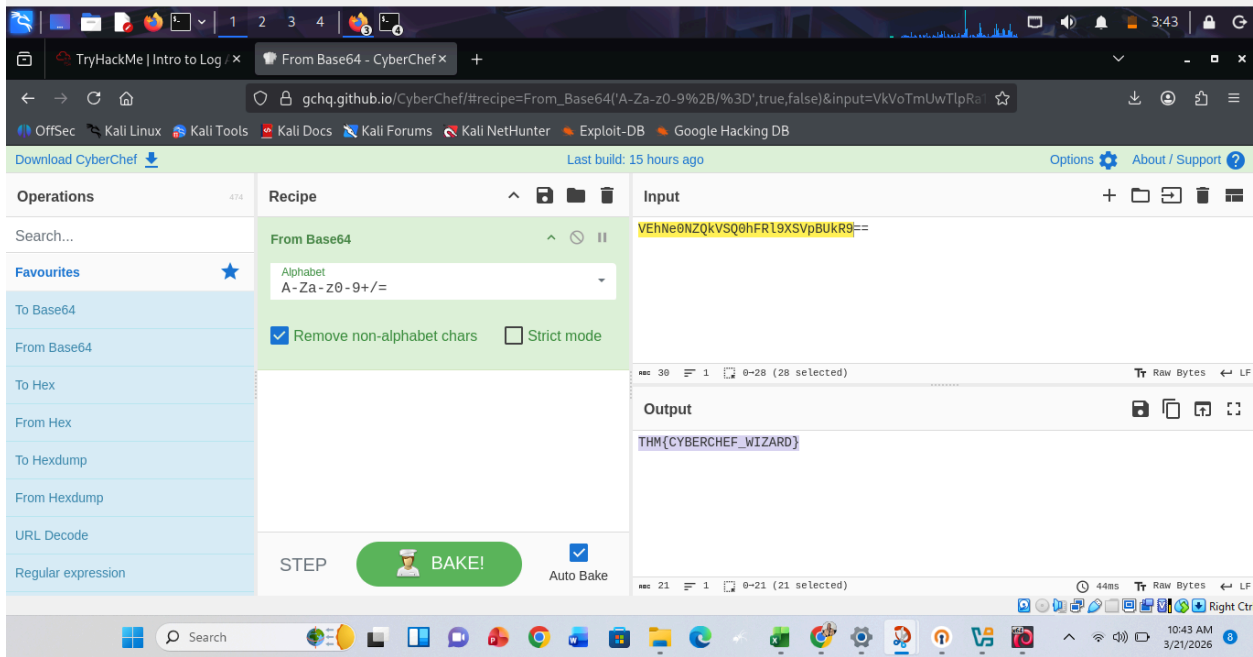
b) Using the same log file from Question #2, a request was made that is encoded in base64. What is the decoded value?

Answer: **THM{CYBERCHEF WIZARD}**

Explanation: i used `grep "GET /" access.log | grep "="` to hunt for the base 64= within GET and POST requests.


```
(root@kali) ~/home/kali/Downloads/task8_logs
(groot@kali) ~/home/kali/Downloads/task8_logs
$ grep "GET" access.log | grep -E "id="
64.62.197.127 - - [27/Sep/2023:04:13:28 +0000] "GET /dns-query?name=dnsscan.shadowserver.org&type=A HTTP/1.1" 401 5196 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/103.0.0.0 Safari/537.36"
83.97.73.87 - - [27/Sep/2023:06:02:06 +0000] "GET /?XDEBUG_SESSION_START=phpstorm HTTP/1.1" 401 5403 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/78.0.3904.108 Safari/537.36"
212.14.17.145 - - [27/Sep/2023:07:00:53 +0000] "GET /VehNe0NZQkVSQ0hFRl9XSvpBUkR9 HTTP/1.1" 401 5196 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:101.0) Gecko/20100101 Firefox/101.0"
43.133.134.155 - - [27/Sep/2023:07:01:06 +0000] "GET /v1wwwsogou?op=8&query=%E7%A8%8F%E5%BB%8A%E9%BE%90%E1%B7%A2 HTTP/1.1" 401 5195 "-" "Mozilla/5.0 (Windows NT 6.1; Trident/7.0; rv:11.0) like Gecko"
```

The encoded value is the string after id=: VEHNe2xvZ19hbmFseXNpc19pcy9mdW59 which i then decoded from base64 and got the flag as the output

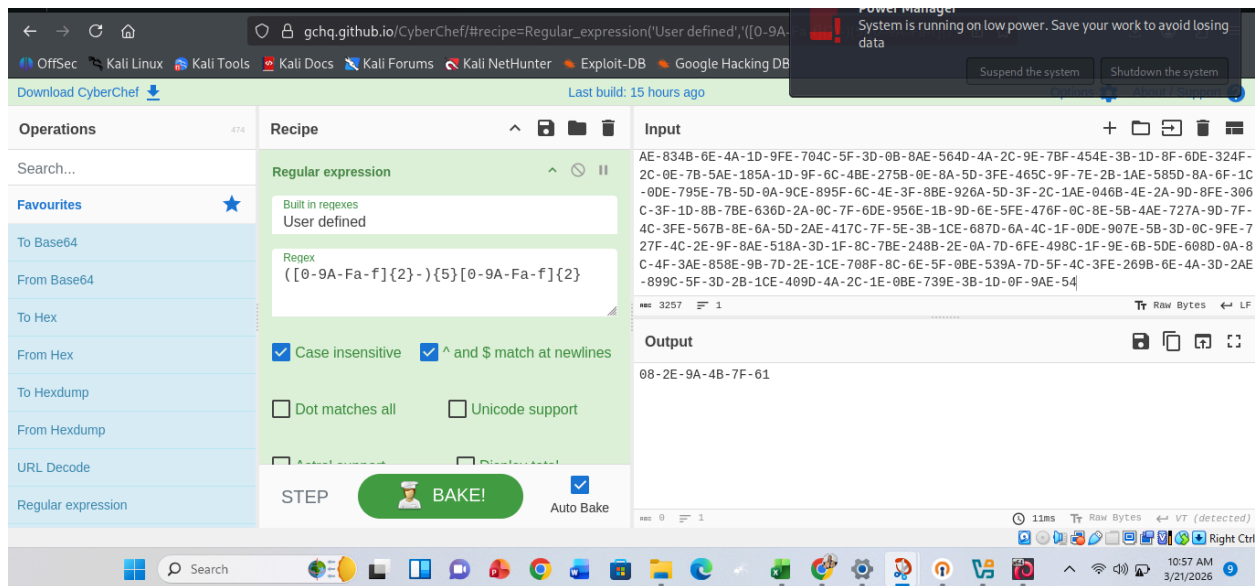


c)Using CyberChef, decode the file named "encodedflag.txt" and use regex to extract by MAC address. What is the extracted value?

Answer:08-2E-9A-4B-7F-61

Explanation:

I had to access the contents of the encoded.txt, i got a long hash.



Below is a screenshot with all the answers.

Locate the "loganalysis.zip" file under `/root/Rooms/introloganalysis/task8` and extract the contents.

No answer needed ✓ Correct Answer

Upload the log file named "access.log" to CyberChef. Use regex to list all of the IP addresses. What is the full IP address beginning in 212?

212.14.17.145 ✓ Correct Answer

Using the same log file from Question #2, a request was made that is encoded in base64. What is the decoded value?

THM{CYBERCHEF_WIZARD} ✓ Correct Answer

Using CyberChef, decode the file named "encodedflag.txt" and use regex to extract by MAC address. What is the extracted value?

08-2E-9A-4B-7F-61 ✓ Correct Answer

9.1 Log analysis tools :Yara and Sigma

What languages does Sigma use?

Answer:**YAML**

What keyword is used to denote the "title" of a Sigma rule?

Answer:**Title**

What keyword is used to denote the "name" of a rule in YARA?

Answer: **Rule**

Answer the questions below

What languages does Sigma use?

YAML

✓ Correct Answer

What keyword is used to denote the "title" of a Sigma rule?

title

✓ Correct Answer

What keyword is used to denote the "name" of a rule in YARA?

rule

✓ Correct Answer

Conclusion and summary

This module showed me that log analysis is really about finding a needle in a digital haystack. It combines the precision of Regular Expressions with the power of the Linux command line. By learning and using the “Golden Pipeline” of grep, sort, and uniq, I’ve moved beyond just reading logs to actually identifying patterns, such as unauthorized access attempts and suspicious encoded payloads.

Using tools like CyberChef to decode Base64 strings and analyze hardware identifiers also helped me understand that being a good analyst isn’t just about finding data, it’s about knowing how to break it down and uncover the real intent behind an attack.

Below is a screen shot showing the module completion

<https://tryhackme.com/room/introtologanalysis/congratulations?step=records>

Share your win with your peers
Your progress can inspire others in your community.

Intro to Log Analysis completed!
marynyambura657 completed another room on their cyber security journey.

Completed tasks
10

Points earned
144

Streak
10

Share on social media
Share your win before starting the next room

in LinkedIn WhatsApp Telegram X Twitter / X f Facebook reddit Reddit

[Go to dashboard →](#)

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